Inferring causality from observational studies: the role of instrumental variable analysis

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Inferring causality from observational studies can be challenging because of the perennial threat of biases from selection, measurement, and confounding. The gold standard study design in clinical research is the randomized controlled trial, because random allocation to treatment ensures that, on average, comparison groups are balanced with respect to both known and unknown prognostic factors. However, most clinically relevant exposure-outcome relationships are not amendable (logistically or ethically) to randomization. Thus, there has been an emergence of analytical approaches over the last several years to improve the validity of inferences made from observational studies. This review presents such an approach, instrumental variable analysis, a technique that has been used by economists for many years but has only recently seen increasing use in the health care literature. This review provides a description of the method, the assumptions underlying it, and recent applications in nephrology outcomes research. A more detailed review of the underlying mathematics, properties of an instrumental variable, and suggested elements for reporting an instrumental variable analysis are provided in the Supplementary Appendix.

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Making causal inferences from data represents a core activity of scientists to gain new insights about the physical world. However, despite our best attempts, we can be led astray by these data if the potential role of systematic and random errors is underappreciated and not properly addressed. Random errors may be diminished with larger sample sizes, but systematic errors frequently persist. The latter are also known as biases and epidemiologists have classified 3 major forms of bias that threaten the validity of epidemiologic studies: selection bias, measurement bias, and confounding bias.

The gold standard study design to mitigate the effects of bias, thus allowing investigators to make causal inferences about treatment effects, is the randomized controlled trial (RCT). However, it is important to emphasize that the RCT design does not exclusively allow investigators to infer causality from data. In fact, both experimental (i.e., RCT) and observational studies can support the development of causally appropriate conclusions. Having said this, it is notable that observational study designs may have a greater susceptibility to bias due to the challenges of creating comparator groups that differ only in the risk factor or treatment of interest, in the absence of randomization.

Conventional multivariable regression models have long been used to analyze patient-level data. However, over the last several years, a number of methodological approaches to improve causal inferences from observational studies have gained prominence in clinical research (e.g., propensity score

Editor's Note

This is the fifth in-depth review of the statistical series. In observational studies, making causal inferences are challenging because of a lack of random exposure assignment. Confounding by indication as well as other unmeasured confounding biases are key challenges when assessing potential treatment effects with an exposure of interests. This article by Fu and Kim explores the concepts of causality, the role of instrumental variables, the validity of their basic assumptions, and its application in clinical nephrology.

methods, directed acyclic graphs, and g-methods). The focus of this review will be an analytical method called instrumental variable (IV) analysis. This approach has been applied by economists for decades to increase the validity of inferences from observational studies but has only recently received increasing attention in the health care literature. A glossary of statistical terms used in this paper is provided in Supplementary Appendix A.

Instrumental variable analysis

IV analysis is a statistical method that increases confidence in causal statements about the effect of a treatment or exposure on an outcome when the treatment or exposure variable is correlated with the error term in a regression equation (also known as endogeneity in the econometrics literature). This correlation arises due to unmeasured confounding, errors in outcome measurement, or reverse causality (i.e., when the outcome affects the exposure level).[1] If left unaddressed, the most commonly used approach for estimating the treatment effect—known as the ordinary least squares (OLS) estimator—is biased in magnitude and/or direction.[1] We present the mathematical basis for these phenomena in Supplementary Appendix B. IV analysis aims to address this type of bias. It was first introduced by Phillip G. Wright in 1928 and has gained popularity among economists since the 1980s.[2–5] The past 2 decades have seen an increase in the use of IV analysis by epidemiologists and the development of various modeling techniques that apply IV analysis to health data.[6–12]

The greatest strength of IV analysis is that it enables causal inference from data without assuming an absence of unmeasured confounders. Thus, it has advantages over approaches such as conventional multivariable regression modeling and propensity score methods as it isolates an independent treatment effect by removing the impact of both measured and unmeasured confounders, under specific conditions.[13] This feature is extremely relevant in observational studies because investigators are unable to control for all potential confounders. Furthermore, IV analysis is useful for enhancing the analysis of an RCT where protocol noncompliance may exist.

The most important task of an IV analysis is to identify a special variable, called an IV, that satisfies the following 3 key assumptions:[14,15]

A1. The IV is strongly associated with the treatment (the *relevance* assumption).

A2. The IV is not directly associated with the outcome, except through its association with treatment (the "exclusion restriction").

A3. The IV is not associated with measured or unmeasured confounders, in other words, the IV does not share common causes with the outcome (the "independence" assumption).

An intuitive way to understand these assumptions is to consider an IV in an observational study as a natural randomizer, which is analogous to a true randomizer in an RCT (Figure 1). An effective randomizer is a cornerstone for a well-executed RCT. When done properly with sufficient sample, the randomizer acts as a proxy for the treatment variable without carrying its associations with confounders. Hence, the randomizer is able to generate variations in the observed outcome that are only attributed to the treatment. To qualify as a proxy, the randomizer must be associated the treatment, which is satisfied by compliance of RCT participants to their allocated treatment. The randomizer must also be independent from all potential confounders and from the outcome (left side of Figure 1).

An observational study lacks a true randomizer (right side of Figure 1). Hence, an IV, as a natural randomizer, mimics the function of a true randomizer if assumptions A1 to A3 are satisfied. Note that A1 corresponds to the perfect compliance assumption, while A2 and A3 establish the effectiveness of IV in reducing alternative pathways between the treatment and the outcome. In some circumstances, additional assumptions beyond the 3 key assumptions for the IV must be considered to ensure proper interpretation of the results. We detail one such assumption—the monotonicity assumption or "no defiers"—in Supplementary Appendix A. This assumption is relevant when the chosen IV can only influence the treatment status of a proportion of patients in the study cohort.

Sources of IVs in health research. To carry out an IV analysis, investigators must first identify an IV using empirical evidence. The key is to ensure A1 and A2 are sufficiently met, that is, the candidate IV is directly associated with the treatment and is not directly associated with the outcome. Obvious violations of A1 and A2 imply immediate rejection of an IV candidate. For example, to confirm a causal relationship between pretransplant dialysis duration and posttransplant mortality among kidney transplant recipients, donor sex does not qualify as an IV due to its lack of association with dialysis duration (i.e., violating A1), while patient age is ineligible because it directly influences mortality (i.e., violating A2). Furthermore, it is preferred that this variable is strongly, rather than weakly, associated with the treatment, as the ability of IV to remove unmeasured confounding bias partly depends on the magnitude of this association (also known as the strength of the IV).[16] Some departures from A3 are to be expected, but, at this stage, they should not be used as the sole reason to reject a potential IV. Several IV candidates have been used in prior health studies and examples are provided in Supplementary Appendix C.[17–19]

IV estimations using 2-stage least square. The most common way to use an IV in regression is through a two-stage least squares procedure (2SLS). The 2SLS is aptly named because it consists of 2 regression equations that are estimated sequentially using the method of OLS. In the first stage, an IV is entered as an independent variable (or as a set of independent variables in the presence of multiple IVs) to predict the treatment variable (X), controlled for a set of measured covariates. In the second stage, the treatment variable values are replaced by the predicted values ($X^\wedge$) from the first stage and entered as an independent variable to predict the outcome (Y), controlled

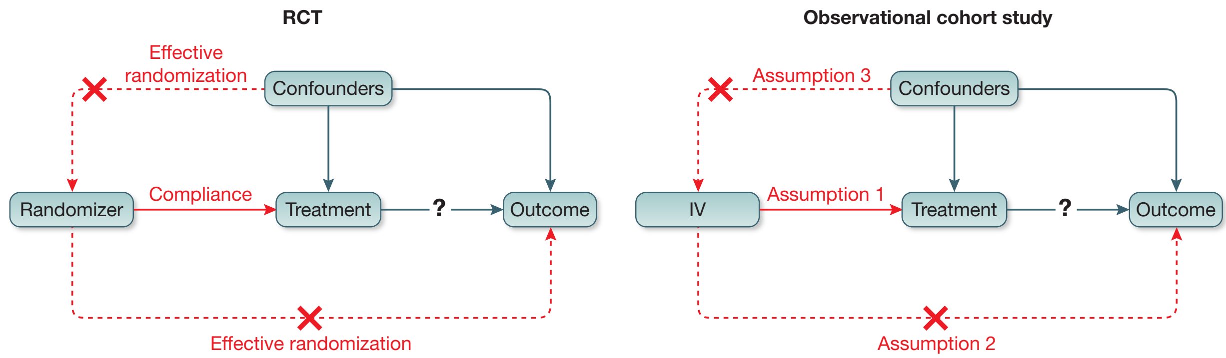

Figure 1 | Causal diagram of a randomized controlled trial (RCT) (left) and an observational study (right). IV, instrumental variable.

for the same set of covariates. The simplest 2SLS assumes both stages describe a linear relationship. The coefficient of the predicted treatment variable ($X^{\wedge}$) in the second stage is the 2SLS-estimator, or the Wald estimator, of the average treatment effect of X on Y.[18] Under the monotonicity assumption, the 2SLS-estimator describes the local average treatment effect of X on Y, which applies exclusively to patients whose exposure or treatment status depends on the value of the chosen IV (Supplementary Appendix A).

The point estimate of the 2SLS-estimator is obtained through the usual OLS method. However, the OLS-based SE is an underestimate because it does not carry the uncertainty from the first-stage regression. Hence, alternative methods including bootstrapping are often used to replace OLS to generate a larger SE, resulting in a wider and more conservative confidence interval (CI) for the 2SLS-estimator.[20]

Common statistical software can be used to perform IV analysis in a 2SLS, including the "ivregress" function in Stata (Stata Corp); the "2sls" option in SAS (SAS Institute) command "PROC SYSLIN"; the "ivreg" function in the R (R Foundation) package "AER: Applied Econometrics with R," and the "IV2SLS" function in the Python (Python Software Foundation) package "linearmodels."

There are many alternative regression methods besides 2SLS that can be used in an IV analysis, such as a 2-stage residual inclusion, a limited information maximum likelihood, and the generalized method of moment.[7,21–23] These methods are beyond the scope of this review. Interested readers are referred elsewhere for a comprehensive review.[18]

Assessing IV assumptions and strength. It is crucial that investigators statistically rule out potential violations of each IV assumption after performing the 2SLS. We summarize some frequently used statistical assessment approaches in Table 1. However, most of these approaches cannot draw definitive conclusions, which necessitates targeted sensitivity analyses that quantify the effect of potential assumption violations on the initial estimation.

For the relevance assumption, investigators must verify that the IV is associated with the treatment variable and assess the strength of this association. A valid IV that is only weakly associated with the treatment can produce biased effect estimates with high variance (i.e., very wide CI), even with large samples.[18] There are 2 statistics that are often used to assess the strength of IV. First, an *F*-statistic >10 for the first-stage equation is a reasonable indicator for a strong IV.[24] Alternatively, one can use the partial r^2 value when adding the IV to the regression model that predicts the treatment variable using the set of observed covariates.[19] A large partial r^2 value implies that a large proportion of variations found in the treatment variable can be explained away by the IV and not by the set of measured covariates. It is worth mentioning that the *F*-statistic is an increasing function of the sample size. Hence, a weak IV can have a large *F*-statistic in large samples.[19] Furthermore, the partial r^2 value does not have a meaningful threshold, which makes it difficult to use in practice.

The exclusion restriction assumption is violated if the IV directly affects the outcome. There is no statistical test that can verify this assumption.[19] However, intuitively evaluating the strength of association between the outcome and the IV may shed light on potential violations of this assumption. After tabulating the outcome variable across IV groups (for categorical IVs) or below or above the mean or median IV (for continuous IVs), a small *P* value may indicate a direct effect from the IV to the outcome. Alternatively, one can regress the outcome variable directly on the IV and the set of measured covariates. A nonzero regression coefficient for the IV (with a small *P* value) may indicate violations of this assumption. However, studies have shown both methods sometimes produce misleading conclusions.[18] Hence, investigators need to rely on subject-matter knowledge. A sensitivity analysis can be performed that adds an interaction term of the treatment variable and the IV to the second-stage equation of 2SLS.[18,19] The coefficient of this interaction term represents the effect modification through the IV on the outcome. By varying this coefficient, investigators can quantify at which threshold the treatment effect is mitigated. We have left the mathematical equations and demonstrations of this sensitivity analysis in Supplementary Appendix D.

Table 1 | Statistical assessments to examine potential violations of the 3 IV assumptions

Assumptions	Statistical assessment methods	Results and implications
A1. The relevance assumption	(1) Perform an F-test with 1 degree of freedom in the first-stage equation. (2) Calculate the partial r^2 when adding the IV to the regression model that predicts the treatment variable using the set of observed covariates.	Having a F-statistic >10 in (1) or a large partial r^2 value in (2) are both reasonable indicators for a strong IV.
A2. The exclusion restriction assumption	Two nonstandard methods: (1) Perform a statistical test on the distribution of the outcome variable across IV groups. (2) Regress the outcome variable on the IV and the set of measured covariates.	Having a small P value in (1), a small P value for the coefficient of the IV in (2), or a nonzero regression coefficient for the IV in (2) reduces the plausibility that A2 is met.
A3. The independence assumption	A nonstandard method is to examine the balance of measured baseline characteristics across IV groups using the standard difference or statistical test.	Balance of measured confounders across IV groups is likely to infer the balance of unmeasured confounders across IV groups.

IV, instrumental variable.

The independence assumption requires an absence of association between the IV and all measured and unmeasured confounders. While it is straightforward to check the balance of measured confounders across levels of IV, no statistical procedures can verify the absence of unmeasured confounders. Some may assume that the balance of measured confounders across IV groups infers balance in unmeasured confounders, but this is speculative at best.[6] When there is clear imbalance of measured confounders by IV, a sensitivity analysis can be performed to quantify the effect of unmeasured confounding.[19] In this analysis, an unmeasured confounder is assumed to be associated with the IV and is independent from the set of measured confounders. We present the mathematical models in Supplementary Appendix E, while an empirical demonstration of this sensitivity analysis can be found elsewhere.[19]

Applications of IV analysis in nephrology outcomes research

Unmeasured confounding is frequently encountered in nephrology outcomes research. Here, we review 2 observational studies that use IV analysis to establish the causal role of certain risk factors on the risk of mortality among patients being treated for chronic kidney disease. Details of the 2 studies are presented in Supplementary Appendixes F and G and can be found in the original papers.[25,26]

Profit status of dialysis centers: using differential distance as an IV. An IV analysis using the 2SLS procedure is seen in Brooks *et al.*[25] who confirmed an absence of the effect of profit status of dialysis centers on 1-year survival of Medicare beneficiaries. In this example, the true survival difference between patients being treated at for-profit versus not-for-profit dialysis centers is confounded by a range of variables, some of which (such as family income) were not captured in the dataset. Hence, a differential distance variable was used as an IV, an idea that appeared previously in a study of acute myocardial infarction.[27] Differential distance was calculated as the straight-line distance from patient's residence at the time of dialysis initiation to the nearest for-profit dialysis center minus the straight-line distance to the nearest not-for-profit center. This distance reflects a patient's relative access to a for-profit center over a not-for-profit center based on the understanding that geographic proximity is likely to influence individual's decision making of care destination (Supplementary Appendix C).

Empirical evidence was first used to check whether differential distance was an appropriate IV based on the 3 key assumptions: it strongly predicted the profit status of the actual dialysis center chosen by the patient, as those who lived closer to a for-profit center tended to go to a for-profit center at a higher rate (A1). A direct impact of differential distance on 1-year mortality was ruled out as there was no apparent connection between the 2 variables (A2). Differential distance is unlikely to be associated with confounders because it is a proximity-based measure that is unrelated to clinical determinants of mortality (A3). However, 2 potential violations of A3 were noted: first, for-profit dialysis centers are often strategically sited in areas with distinct socioeconomic and clinical composition of residents; furthermore, the use of differential distance assumes that individual's decision on residence location precedes dialysis initiation (i.e., patients would choose dialysis center with respect to their current residence location). However, it does not consider patients who have purposefully relocated to live closer to their preferred dialysis center.

A 2SLS procedure was carried out wherein differential distance was operationalized as a set of 4 indicator variables to denote the 5 quintiles (every 20th percentile). A linear probability model was used for both stages.[28] At the first stage, the for-profit status of patient's dialysis center was predicted by the 4 differential distance indicator variables and the set of measured confounders. The predicted probability of initiating dialysis at a for-profit center from the first-stage model was then used to predict 1-year mortality with the same set of measured confounders in the second-stage model. The IV-estimated effect of starting dialysis at a for-profit center on mortality was not significant (coefficient = 0.00248; SE = 0.00754; P = 0.746). On the contrary, a direct linear probability model suggested for-profit centers to be associated with lower mortality risk, presumably because of unmeasured confounders.

IV assumptions were statistically assessed to establish the validity of these results. The *F*-statistics (7909.59) provided evidence that the 4 differential distance indicator variables were strong IVs (A1). A Hausman test found that excluding the 4 differential distance indicator variables from the second-stage equation was appropriate ($P = 0.5877$), indicating a lack of direct statistical association between mortality and differential distance (A2). Measured confounders were not quite balanced below or above the median differential distance (−3.51 miles); however, no specific procedures were conducted to address this concern given the adjustment of these covariates in the 2SLS (A3).

Dialysis duration before kidney transplant: using patient blood type as an IV. Survival data that describe the time-to-event are common in clinical outcomes studies. Due to censoring, regression models are often replaced by survival models to reduce bias. There are several ways to implement IVs with survival data. Here we focus on a straightforward adaption of 2SLS where a Cox proportional hazards model is used in the second stage.[8] Interested readers are referred elsewhere for the use of IV in Cox models with nonproportional hazards and in Aalen additive hazards models.[9,11]

A recent study by our group tested the causal relation between the duration of pretransplant dialysis and the risk of posttransplant mortality among 4440 kidney transplant recipients who were followed until death or at least 1 year after transplantation.[26] Unmeasured confounding is a concern as patients may be subjected to prolonged dialysis for reasons that also affect their survival. For example, investigators were unable to identify the expanded criteria donor (ECD) status of the transplanted kidney. Patients who consent to receive an ECD kidney spend less time on dialysis awaiting transplantation when compared to others in their age group, but are also known to have reduced survival after transplantation when compared to non-ECD transplants.[29] If left unaddressed, ECD status would lead to an underestimation of the true dialysis effect, as shorter dialysis time would appear to increase the risk of mortality.

To mitigate the effects of unmeasured confounding, patient blood type was used as an IV, an idea originating from a prior report.[30] Patients with blood type AB generally have shorter waiting times until transplant (i.e., shorter pretransplant dialysis time) compared to patients with other blood types. Hence, patient blood type strongly predicts dialysis duration (A1). Moreover, blood type does not directly influence kidney transplant outcomes (A2). Finally, blood type is unlikely to be associated with confounders of dialysis durations and transplant mortality, except through 1 potentially problematic variable, patient race. Because race is known to differ by blood type and influence both wait time on dialysis and transplant outcomes, it may confound the effect estimation in the IV analysis (A3).

Three dummy variables were created to represent patient blood type, indicating A, B, and AB, respectively (O was the reference blood type). The first-stage regression of the 2SLS predicted the natural log of dialysis duration using the 3 blood type dummy variables and the set of measured covariates. The second-stage regression was a threshold-adjusted Cox proportional hazards model that predicted the mortality hazard using the predicted dialysis duration from the first-stage regression and the same set of covariates.[8,31,32] Detailed descriptions of this model are discussed in Supplementary Appendix G. A 1-stage threshold-adjusted Cox proportional hazards model was also estimated to compare the results with the IV analysis.

In this example, the 2 models yielded similar results regarding the threshold of dialysis duration and the dialysis effect. Both models identified 3 years after dialysis initiation as a significant threshold. Before 3 years on dialysis, the mortality hazard ratio (HR) was about 42% (HR, 1.42; 95% CI, 1.03, 1.55) per year in the IV analysis and was slightly higher at 46% (HR, 1.46; 95% CI, 1.25,1.71) in the 1-stage analysis. After the third year, the yearly dialysis effect on mortality was reduced to about 5% in both IV and 1-stage analysis. Hence, these findings suggest that while the risk of posttransplant mortality continues to grow with longer dialysis treatment, there might exist a dialysis threshold after which dialysis heightens mortality less than it did before that threshold. These results underscore efforts to shorten the overall dialysis duration before transplantation and support tailored interventions to ensure the safety and well-being of patients during the early stages of dialysis.

The use of patient blood type as an IV was statistically assessed. Its strength was verified by a large *F*-statistic that exceeded 10 (A1). The mortality outcome was balanced across blood type groups in the baseline table (A2). As expected, measured confounders were balanced across blood type groups, except for patient race ($P < 0.001$). A subgroup analysis stratified by race was thus performed to rule out the concern for differential dialysis effects by race. Because the results were similar to the initial effect estimated for all patients, the IV analysis generated results that were robust to a potential violation of A3 by patient race.

Conclusions

There are a number of statistical tools that can be used to improve validity of inferences made from observational studies. This review focused on the IV analysis method, which has had a long history in economics literature. When applied correctly, valid estimates of the causal relationship between an exposure (or treatment) and outcome can be determined. However, the assumptions that underlie this method can be difficult to assess and may not be testable with the available data. Moreover, the estimates from an IV analysis should technically be applied only to patients whose exposure or treatment status depends on the instrument (i.e., patients who are "compliers" or "marginal"). With an awareness of these limitations, and its judicious application to data when a reasonable IV is identified, IV analysis is a useful addition to the statistical tool kit of the clinical researcher looking to improve causal inferences from observational studies. For the interested reader, a general approach to presenting the results of an IV analysis is provided in Supplementary Appendix H.

DISCLOSURE

All the authors declared no competing interests.

SUPPLEMENTARY MATERIAL

Supplementary File (PDF)

Supplementary Appendix A. Glossary.

Supplementary Appendix B. Bias of OLS-estimator when the treatment variable is correlated with the error term.

Supplementary Appendix C. Examples of instrumental variables in health research.

Supplementary Appendix D. Sensitivity analysis on the violations of A2 (exclusion restriction assumption).

Supplementary Appendix E. Sensitivity analysis on the violation of A3 (the independence assumption).

Supplementary Appendix F. Using differential distance as an IV to estimate a causal relationship between profit status of dialysis center and 1-year mortality.

Supplementary Appendix G. Using blood type as an IV to estimate a causal relationship between longer pretransplant dialysis duration (years) and mortality following kidney transplantation.

Supplementary Appendix H. Suggested items to report in an IV analysis.

Supplementary Appendix References.

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